Sri Lanka Institute of Information Technology

Faculty of Computing — Department of Computer Systems Engineering

IE3122 – Network Security

Week 2 Tutorial Worksheet: Networking and Security Revision

Field	Detail
Based on	Week 2 — Networking and Security Revision lecture (key terms, threat modeling, hacker types, attack categories, security controls)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO3 (analyse network/communication requirements)
Format	13 structured questions, no MCQs — answer in your own words
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic — use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's lecture. They're not just recall — several ask you to explain a mechanism in your own words or connect it to a security implication. Write full answers, not one-line definitions: a good answer should read like a study note you could revise from later without needing to reopen the slides.

Key Terms & Threat Modeling

Q1. Define asset, vulnerability, threat, exploit, and risk in your own words, and explain how the five relate to each other using one connected example. *[LO1]*

Q2. A university keeps student records on a server with an outdated, unpatched operating system. Apply the threat-modeling process (asset → threat → vulnerability → risk) to this situation, and state whether you'd rate the risk low, medium, or high — with justification. *[LO3]*

How Networking and Threats Have Evolved

Q3. Explain how the growth of cloud services, mobile devices, and IoT has expanded the attack surface compared to the isolated-LAN era. Give one concrete example of a threat that simply didn't exist before this shift. *[LO1]*

Types of Hackers & Motivations

Q4. Compare white hat, black hat, and grey hat hackers. Then explain what distinguishes blue hat and red hat hackers from those three. *[LO1]*

Q5. A group breaks into a government agency's systems and publicly leaks documents to protest a policy, without seeking any payment. Identify the most likely threat-actor category and motivation, and explain your reasoning. *[LO1, LO3]*

Malware

Q6. Explain the key difference in how a virus, a worm, and a trojan each spread. Why does that difference matter when deciding how to defend against each one? *[LO1]*

Q7. Describe what ransomware does and why it's considered especially damaging to an organisation compared to, say, spyware. Reference a real example if you can. *[LO1]*

Social Engineering

Q8. Describe phishing, pretexting, and baiting, making clear what distinguishes each from the others. *[LO1]*

Q9. Your company's help desk receives a call from someone claiming to be a senior manager, urgently requesting a password reset without normal verification. Which social-engineering technique does this resemble, and what policy would you recommend to prevent it succeeding? *[LO3]*

DoS & DDoS

Q10. Explain the difference between a DoS and a DDoS attack, and why a botnet makes a DDoS attack significantly harder to defend against than a DoS attack. *[LO1]*

Access & Reconnaissance

Q11. Explain the difference between a reconnaissance attack and an access attack, and why reconnaissance usually happens first in a real intrusion. *[LO1]*

Security Controls & Layered Defense

Q12. Describe administrative, technical, and physical security controls, giving one concrete example of each. *[LO1]*

Q13. A small clinic (see this week's lab scenario) has only one control in place: a firewall on its Internet connection. Explain, referencing at least two of today's attack categories, why this single control leaves them exposed — and recommend one additional control from a different category. *[LO3]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. For every attack category covered this week, note which later week covers its full technical defence in detail (e.g. Firewalls in Week 7, AAA in Week 8) — link this content forward as the module progresses.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1–LO2) and final exam (MCQ + structured essay, LO1–LO5) — the scenario-style questions above are close in format to the structured-essay section.